

Human Resource Management (HRM) System

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1. Introduction

- **Problem:** small and medium organizations still track attendance on paper or spreadsheets, handle leave by e-mail, and calculate salaries by hand – slow, error-prone and opaque to employees.
- **Solution:** a web-based, **multi-company HRM System** that automates attendance, leave and monthly payroll with role-based access.
- **Roles:** Super Admin (onboards companies), Company Admin (manages own company), Common employee (checks in/out, applies for leave, views own records).
- **Automation:** a daily job marks absent employees; a monthly job generates the previous month's payroll.
- **AI HR Assistant:** answers HR questions in natural language from the company's own data.

2. Objectives

- **Broad objective**
 - Design and develop a secure, web-based, multi-company HRM system that automates attendance, leave and payroll administration.
- **Specific objectives**
 - Role-based access control scoped to the user's own company
 - Check-in/check-out with automatic late and early-leave detection; scheduled absent marking
 - Leave workflow: request → approve/reject → reflected in attendance
 - Automatic monthly payroll with deduction rules; live payroll projection
 - Daily, monthly and per-employee attendance analytics
 - AI assistant using read-only, company-scoped data tools
 - JWT authentication, refresh-token rotation, BCrypt password hashing

3. Methodology and Process Model

- **Process model:** Agile – iterative and incremental with feedback at the end of every iteration.
- **Iterations (12–13 weeks)**
 - Sprint 0 – requirements, Clean Architecture skeleton, database schema
 - Sprint 1 – JWT authentication, company / role / department management
 - Sprint 2 – employee management, pagination, Swagger security
 - Sprint 3 – attendance, automatic absent marking, leave workflow
 - Sprint 4 – payroll job, live payroll status, AI assistant
 - Final – Angular integration, testing, documentation
- **Why Agile:** requirements were refined from working software (e.g. the check-in rule after auto-absent marking), modules could be demonstrated early, defects found early.
- **Stack:** Angular · ASP.NET Core Web API (.NET 10) · Clean Architecture + CQRS (MediatR) · EF Core · SQL Server · Microsoft.Extensions.AI.

Organization Overview

- **Organization:** AKIJ iBOS Ltd., Dhaka, Bangladesh
- **Services:** Provides AI-driven software, enterprise solutions, and full-stack digital services designed for modern businesses
- **Vision / Mission:** Vision is to build smart systems
- **My position:** Backend Intern – analysis, design and full-stack development of the HRM System
- **Organizational supervisor:** Saidur Rahman Akash, Software Engineer L-2
- **Duration:** 12 weeks (1st June, 2026 – 31st August, 2026)



5. System Requirements

User & system requirements – each user need with the system requirements derived from it (report §3.1.1).

- **1. Users need to securely access the system according to their assigned roles.**
 - The system needs to allow registered users to log in using their e-mail address and password.
 - The system needs to issue a short-lived access token and a refresh token on successful login.
 - The system needs to renew the access token silently using the refresh token without asking for the password again.
 - The system needs to restrict every API endpoint and every page according to the user's role.
 - The system needs to scope all company data to the company of the logged-in user.
 - The system needs to allow authenticated users to log out and revoke their refresh token.
- **2. The Super Admin needs to manage the companies that use the platform.**
 - The system needs to allow the Super Admin to create and update company profiles (name, e-mail, phone, address, logo, subscription plan).
 - The system needs to allow companies to be activated or deactivated.
 - The system needs to list all companies with pagination and view a single company's details.
- **3. The Company Admin needs to manage departments.**
 - The system needs to allow departments to be created and updated under the admin's company.
 - The system needs to display the number of employees in each department.
 - The system needs to list departments with pagination and sorting.

5. System Requirements (Cont.)

- **4. The Company Admin needs to manage roles and employee profiles.**
 - The system needs to allow roles (e.g. Company Admin, Common) to be created and updated.
 - The system needs to allow employee profiles to be created and updated with code, name, contact details, department, role, join date, salary and status.
 - The system needs to store passwords only as salted hashes.
 - The system needs to allow employees to be activated or deactivated.
 - The system needs to list employees of a company, optionally filtered by department, with pagination.
- **5. Employees need to record daily attendance.**
 - The system needs to allow an employee to check in; the first check-in of the day creates the attendance record.
 - The system needs to allow the same employee to check out later in the day and calculate working hours.
 - The system needs to classify a check-in as Present or Late against the configured shift start time and record the minutes late.
 - The system needs to record early leave minutes when check-out happens before the shift end.
 - The system needs to prevent a second check-out for the same day.
- **6. The system needs to mark absent employees automatically.**
 - The system needs to run a scheduled job every working day at a configured time (9:30 AM Bangladesh time) that marks every active employee without an attendance record as Absent.
 - The job needs to skip weekly holidays and employees on approved leave.
 - If the server was not running at the scheduled time, the job needs to catch up when the server starts.
 - A late employee who checks in after being auto-marked Absent needs to have the record converted to Late rather than being blocked.



5. System Requirements (Cont.)

- **7. Employees and administrators need to manage leave.**
 - The system needs to allow an employee to submit a leave request with type, date range and reason; total days are calculated by the system.
 - The system needs to allow a Company Admin to approve, reject or cancel a leave request and record who approved it and when.
 - The system needs to reflect approved leave in attendance as “On Leave” for each working day in the range.
 - The system needs to list leave requests by status and show an employee’s leave history.
- **8. The system needs to generate payroll automatically.**
 - The system needs to generate the payroll of the previous month for all active employees on the first day of each month.
 - The payroll needs to deduct a fixed penalty for each absence and convert every three late arrivals into one absence.
 - The system needs to guarantee that a month’s payroll is generated only once, even if the server restarts.
 - The system needs to allow a Company Admin to view the payroll of an employee or of the whole company for a month.
 - The system needs to show a live, non-stored projection of the current month’s payroll for an employee.
- **9. The Company Admin needs attendance analytics.**
 - The system needs to provide a daily summary (present, late, on leave, absent) for the company.
 - The system needs to provide a monthly summary with average attendance rate, late arrivals, perfect attendance, most punctual department and highest absentee(s), reporting all names in case of a tie.
 - The system needs to provide per-employee monthly statistics and day-by-day records.

5. System Requirements (Cont.)

- **10. HR staff need to ask questions in natural language.**
 - The system needs to provide an AI assistant endpoint that accepts a question and returns an answer based only on the company's own attendance, leave and employee data.
 - The assistant needs to be able to resolve an employee's name to their record before answering employee-specific questions.
 - The assistant needs to answer in the language the question was asked in.
 - The assistant needs to state clearly when a question cannot be answered from the available data.
 - The response needs to include which data tools were used so the answer can be trusted and audited.
- **11. The system needs to expose a documented API for the frontend.**
 - The system needs to publish an OpenAPI (Swagger) description of every endpoint with bearer-token support for testing.
 - The system needs to return consistent paginated responses (items, page number, page size, total count, total pages).
 - The system needs to return meaningful validation errors and HTTP status codes.



5. Functional Requirements

- FR-01: The system shall allow registered users to log in with e-mail and password, refresh their session and log out securely.
- FR-02: The system shall enforce different permissions for Super Admin, Company Admin and Common users on every endpoint.
- FR-03: The system shall scope every query and command to the company identified in the caller's token.
- FR-04: The system shall allow the Super Admin to create, update, activate/deactivate and list companies.
- FR-05: The system shall allow the Company Admin to create, update and list departments with employee counts.
- FR-06: The system shall allow the Company Admin to create, update and list roles.
- FR-07: The system shall allow the Company Admin to create, update, activate/deactivate, search and list employees with pagination.
- FR-08: The system shall record employee check-in and check-out, computing status, late minutes, early-leave minutes and working hours.
- FR-09: The system shall automatically mark absent employees every working day and catch up on missed runs at startup.



5. Functional Requirements (Cont.)

- FR-10: The system shall allow employees to submit leave requests and administrators to approve, reject or cancel them.
- FR-11: The system shall convert approved leave into “On Leave” attendance records for the affected working days.
- FR-12: The system shall generate the previous month’s payroll automatically on the first day of each month, exactly once.

5. Non-Functional Requirements

- NFR-01 (Security): The system shall use JWT authentication with 15-minute access tokens, HttpOnly/Secure/SameSite refresh-token cookies, BCrypt password hashing and role-based authorization on every endpoint.
- NFR-02 (Data isolation): No request shall be able to read or modify data of a company other than the caller's own company.
- NFR-03 (Performance): List endpoints shall be paginated (default 50, maximum 100 items) and typical API responses shall be returned within one second; AI answers within ten seconds.
- NFR-04 (Reliability): Scheduled jobs shall be idempotent and shall recover from missed runs when the server restarts.

6. Use Case Diagram

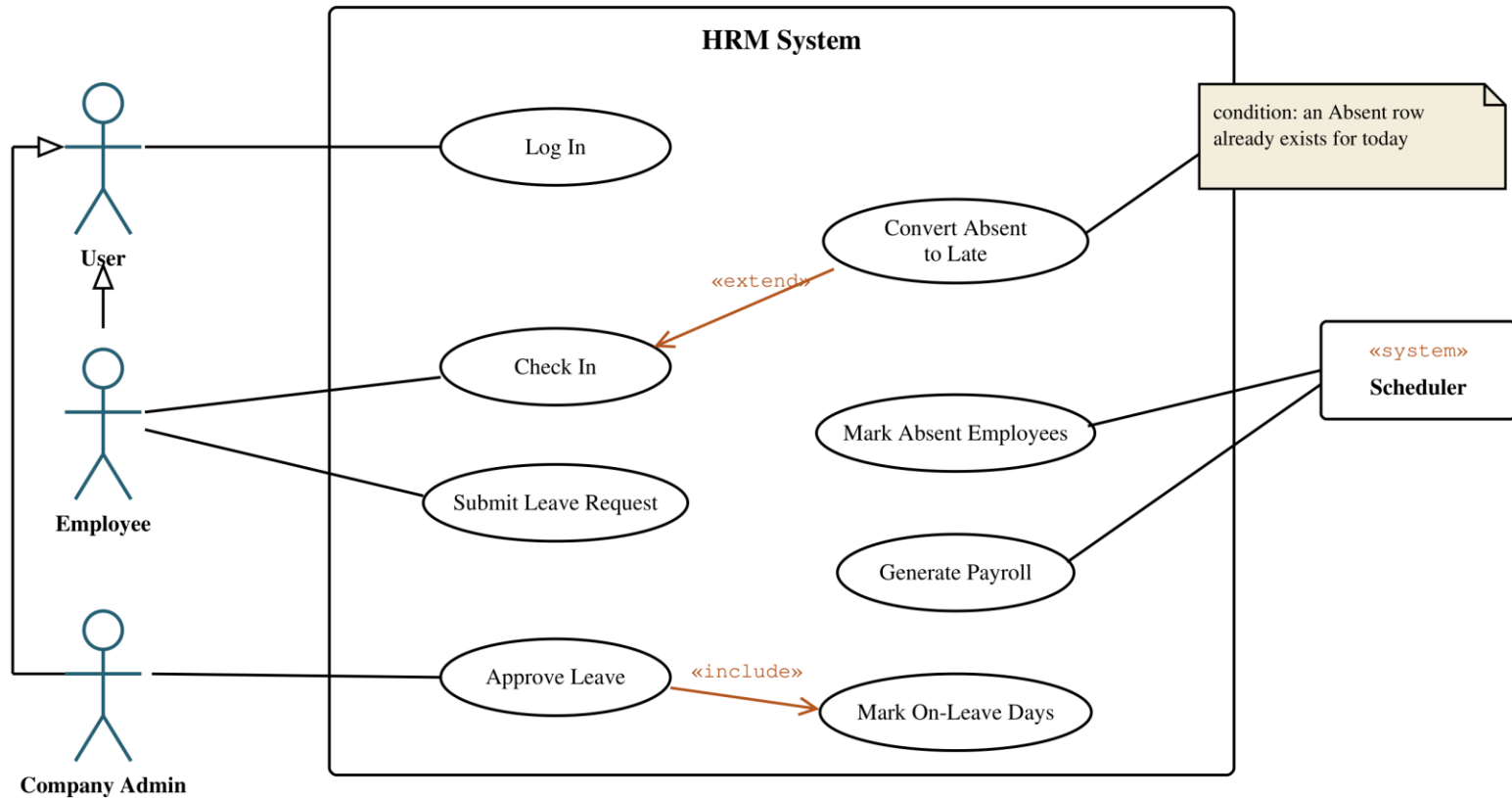


Figure: Use case diagram of the HRM System – Employee and Company Admin generalize User; Scheduler is a secondary «system» actor; Approve Leave «includes» Mark On-Leave Days; Convert Absent to Late «extends» Check In when an Absent row already exists.

7. Database Design

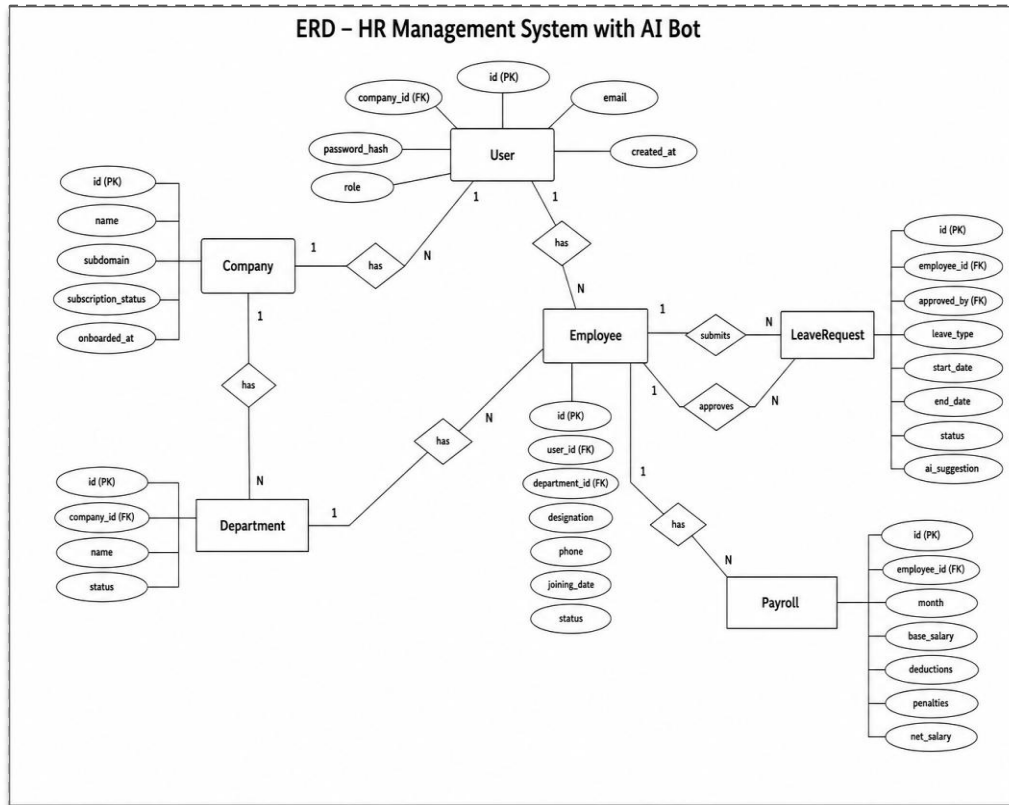


Table	Purpose
Company	Tenant organizations
Department, Role	Company structure, authorization roles
Employee	Users: profile, credentials, salary
Attendance	One row per employee per day
LeaveType, LeaveRequest	Leave workflow
Payroll	Monthly salary with deductions
RefreshToken	Session renewal / revocation
AiConversation, AiAuditLog, AiReport	AI usage and audit

- 12 normalized tables; every business table carries **CompanyId** for tenant isolation.
- Attendance status: Present / Late / Absent / On Leave; Payroll: Basic – AbsentDeduction – LateDeduction = Net.

8. Implementation – Screenshots

Log In Page

Dashboard

My Company
View and edit your company's details.
[Go to My Company](#)

Departments
Manage departments per company.
[Go to Departments](#)

Employees
Browse and manage employee records.
[Go to Employees](#)

Leave Approvals
Review pending leave requests.
[Go to Approvals](#)

Attendance
View attendance for a company and date.
[Go to Attendance](#)

Payroll
View payroll across a company.
[Go to Payroll](#)

Employees

Department: All departments

Name	Email	Role	Status
Mash Tarafder	mashrafeubat@gmail.com	Super Admin	Active
Sahadat Sanbid	sahadat@s.s	Company Admin	Alive
Din Islam	r@r.r	Common	Alive
Mash T	mash@m.m	Common	Active
rakin a	r@r.r	Common	Active

Attendance

Today
Check-in: —
Check-out: —
[Check In](#) [Check Out](#)

This Month
2 Present, 2 Late, 0 Leave, 20% Ratio

Date	Check In	Check Out	Status
2026-07-18	18:23:14.7323355	—	Late
2026-07-17	18:27:28.3173534	—	Late
2026-07-16	00:55:54.2961033	10:34:01.2765384	Present
2026-07-20	—	—	Absent
2026-09-04	22:40:06.9885406	22:40:20.4430826	Late

Angular frontend consuming the Web API; role decides the dashboard after login.



8. Implementation – Screenshots (Cont.)

Sahadat - Company Admin

My Leave Requests

+ New Request

From	To	Days	Reason	Status
2026-09-15T00:00:00	2026-09-20T00:00:00	6	123456	Approved

Sahadat - Company Admin

Attendance — Monthly Summary

Day Month

Department: All departments, Month: September, Year: 2026

9%
Avg. Attendance

2
Late Arrivals

0
Perfect Attendance

HR
Most Punctual Dept.

Highest absentee this month: Din Islam, Mash T, Mash Tarafder, rakin a - Late rate: 9.09%

Employee	Department	Total Absent
Din Islam	HR	3
Mash T	HR	3
Mash Tarafder	HR	3
rakin a	HR	3
Sahadat Sanbid	HR	2

Sahadat - Company Admin

Payroll — Company

Month: 9, Year: 2026

Employee	Employee Id	Basic Salary	Absent Deduction	Late Deduction	Net Salary
No payroll records for this period.					

Live Estimate (Current Month)

Computed on demand from this month's attendance so far — not a finalized/generated payroll record.

Employee		Basic Salary	Absent Deduction	Late Deduction	Net Salary
Mash Tarafder	View Live Estimate	7000	300	0	6700
Sahadat Sanbid	View Live Estimate				
Din Islam	View Live Estimate				

HR Assistant

Ask about attendance and leave in plain language. Each question is answered on its own.

You
Who was absent today?

Assistant
Absent today (2026-09-14):
- Mash Tarafder
- Sahadat Sanbid
- Din Islam

0 / 1000

Send

Payroll rule: 100 BDT per absence; every 3 late arrivals = 1 absence. AI answers list the data tools used.

9. Testing

- **Approach:** black-box (Swagger + Angular UI), white-box (handlers, branches), grey-box (DB verification); unit → integration → system → acceptance.

Scenario	Expected	Result
Login and token refresh	Token with Role & CompanyId; refresh only with valid cookie	Pass
Role-based, company-scoped access	401 without token, 403 wrong role, no cross-company data	Pass
Check-in after auto-absent marking	Absent row converted to Late, no duplicate	Pass (after fix)
Payroll idempotency	Exactly one row per employee per month	Pass
AI question “Who was late the most?”	Correct name + count; toolsUsed listed	Pass

- **Non-functional:** 15-min token expiry, cookie flags, pagination limits, AI response time (2–10 s), Windows/Linux portability.

10. Limitations and Future Plan

- **Limitations**

- Shift times, holidays and penalty are global constants, not per-company settings
- Payroll covers basic salary and deductions only; generated globally
- AI assistant: attendance and leave only, no payroll, no conversation memory
- No notifications; no dedicated mobile app

- **Future plan**

- Per-company configuration of shifts, holidays, leave balances and deductions
- Allowances, tax, bonus and payslip export (PDF/Excel)
- AI: payroll tools, memory, proactive monthly reports
- E-mail/push notifications; mobile app with location/biometric check-in
- Automated test suites with CI/CD

11. Conclusion

- Delivered a working, secure, multi-company HRM System: attendance, leave, payroll, analytics and an AI assistant.
- Automation removes manual registers and hand-calculated salaries; company-level isolation is enforced by design.
- Applied the full lifecycle – requirements, Clean Architecture/CQRS design, FP estimation, RMMM, testing.
- Practicum value: hands-on experience with Angular, .NET 10, EF Core, JWT security, background services and LLM tool-calling, plus documentation and communication skills.
- The system is a solid base for the per-company configuration and payroll extensions planned next.

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Any Questions ?

THANK YOU